



MASTER-KIDS · OLYMPIAD MOCK SERIES · CLASS 4

Mathematics Olympiad Mock Paper 81

Olympiad Mock — IMO Level 1 Pattern · 35 Questions

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Olympiad Mock (IMO Level 1 Pattern)	Topic Focus	Full-length Olympiad paper
Questions	35	Time Allowed	60 minutes
Total Marks	40	Worksheet No.	81 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

This is Worksheet 81 — the first of the Olympiad Mock papers in the Class 4 Mathematics series. It follows the exact pattern of the SOF International Mathematics Olympiad (IMO) for Class 4: four sections — Logical Reasoning, Mathematical Reasoning, Everyday Mathematics, and a high-scoring Achievers Section worth 2 marks per question (40 marks in all). The questions suit children from CBSE, ICSE and State Board backgrounds alike. Sitting this paper under timed conditions is the best way to prepare a child for the real exam: it builds speed, accuracy and the habit of attempting the tougher 2-mark questions last.

Learning Objectives

- Experience the real four-section structure and timing of a Class 4 Maths Olympiad.
- Apply logical-reasoning skills: series, coding, ranking, direction, analogy and figure patterns.
- Demonstrate mathematical reasoning across numbers, fractions, geometry and measurement.
- Solve everyday word problems involving money, time, weight, capacity and data.
- Attempt high-value Achievers questions that need multi-step thinking.

Instructions

1. This is a full-length Olympiad-pattern paper with 35 questions across four sections.
2. All sections carry 1 mark per question, EXCEPT the Achievers Section, where each question carries 2 marks — these are the hardest, high-reward questions, exactly like a real SOF Olympiad.
3. Each question has one correct option: (A), (B), (C) or (D). All questions are compulsory and there is no negative marking.
4. Suggested time: 60 minutes. Try to finish the first three sections in about 40 minutes, leaving 20 minutes for the Achievers Section.
5. Use of a calculator is not permitted. Do rough work on a separate sheet.
6. Check answers against the Answer Key after finishing, then study every Detailed Solution.

Marking scheme: Sections 1-3: +1 mark each (30 questions = 30 marks) · Achievers Section: +2 marks each (5 questions = 10 marks) · Maximum 40 marks · No negative marking.



SECTION 1 · LOGICAL REASONING · Q1-Q10 · (1 mark each)

Choose the correct option for each question.

Q1. Find the next number in the series: **2, 6, 12, 20, 30, __**

- (A) 40 (B) 42
(C) 44 (D) 36

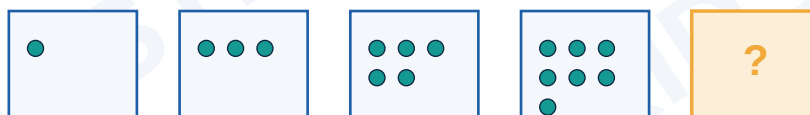
Q2. If **MONDAY** is coded as **NPOEBZ** (each letter moved one step forward), how is **FRIDAY** coded?

- (A) GSJEBZ (B) GSJDBZ
(C) ESJEBZ (D) GSIEBZ

Q3. Pointing to a photo, Aarti says, 'He is the son of my mother's only son.' How is the boy in the photo related to Aarti?

- (A) Her son (B) Her brother
(C) Her nephew (D) Her cousin

Q4. Which figure comes next in the pattern? (Count the dots in each box.)



- (A) 8 dots (B) 9 dots
(C) 10 dots (D) 11 dots

Q5. Reena is taller than Sita but shorter than Meena. Gita is taller than Meena. Who is the TALLEST?

- (A) Reena (B) Sita
(C) Meena (D) Gita

Q6. Find the odd one out: **square, rectangle, triangle, circle.**

- (A) square (B) rectangle
(C) triangle (D) circle

Q7. A clock shows 3 o'clock. What is the smaller angle between the hour hand and the minute hand?

- (A) 45° (B) 60°
(C) 90° (D) 120°

Q8. Complete the analogy: **2 is to 8 as 3 is to __** (each first number is multiplied by the same number).

- (A) 9 (B) 12
(C) 18 (D) 27



Q9. Ravi walks 4 km North, then turns right and walks 3 km. In which direction is he now facing, and how far is he from the start in a straight line?

- (A) Facing East, 5 km away (B) Facing East, 7 km away
(C) Facing West, 5 km away (D) Facing South, 1 km away

Q10. In a row of 20 children, Anu is 7th from the left. What is her position from the right?

- (A) 12th (B) 13th
(C) 14th (D) 11th

SECTION 2 · MATHEMATICAL REASONING · Q11-Q20 · (1 mark each)

Q11. What is the place value of **7** in the number **47,365**?

- (A) 7 (B) 70
(C) 700 (D) 7,000

Q12. Which of these is the correct expanded form of **6,408**?

- (A) 6,000 + 400 + 8 (B) 6,000 + 40 + 8
(C) 600 + 40 + 8 (D) 6,000 + 400 + 80

Q13. What is **$72 \div 8 + 5 \times 3$** ?

- (A) 24 (B) 33
(C) 9 (D) 39

Q14. Which fraction is equal to **$\frac{1}{2}$** ?

- (A) $\frac{2}{3}$ (B) $\frac{3}{6}$
(C) $\frac{2}{5}$ (D) $\frac{4}{6}$

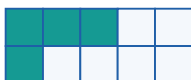
Q15. How many lines of symmetry does a **square** have?

- (A) 1 (B) 2
(C) 4 (D) 0

Q16. Find the missing number: **$8 \times \underline{\quad} = 56$**

- (A) 6 (B) 7
(C) 8 (D) 9

Q17. The shaded part of the grid shows a fraction. Each small square is equal. What fraction is shaded?



- (A) $\frac{4}{10}$ (B) $\frac{3}{10}$
(C) $\frac{5}{10}$ (D) $\frac{4}{8}$



Q18. What is the perimeter of a square whose side is 9 cm?

- (A) 18 cm (B) 27 cm
(C) 36 cm (D) 81 cm

Q19. Round **3,567** to the nearest thousand.

- (A) 3,000 (B) 3,500
(C) 4,000 (D) 3,600

Q20. Which number is both a multiple of 3 and an even number?

- (A) 9 (B) 15
(C) 18 (D) 25

SECTION 3 · EVERYDAY MATHEMATICS · Q21-Q30 · (1 mark each)

Q21. A pencil costs ₹6. How much will 1 dozen pencils cost?

- (A) ₹60 (B) ₹66
(C) ₹72 (D) ₹78

Q22. A water tank holds 250 litres. It is $\frac{4}{5}$ full. How many litres of water are in the tank?

- (A) 150 L (B) 200 L
(C) 210 L (D) 50 L

Q23. A train leaves at 9:45 a.m. and reaches its stop at 1:15 p.m. How long is the journey?

- (A) 3 hours (B) 3 hours 30 minutes
(C) 4 hours (D) 3 hours 15 minutes

Q24. Meena had ₹500. She bought a book for ₹185 and a pen for ₹45. How much money is left with her?

- (A) ₹270 (B) ₹260
(C) ₹280 (D) ₹230

Q25. A garden is 12 m long and 8 m wide. What length of fencing is needed to go all around it once?

- (A) 20 m (B) 40 m
(C) 96 m (D) 48 m

Q26. There are 6 rows of chairs with 15 chairs in each row. If 18 chairs are empty, how many chairs are occupied?

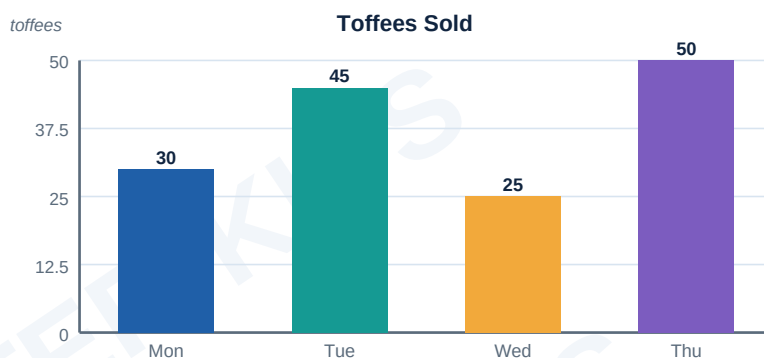
- (A) 72 (B) 90
(C) 108 (D) 70

Q27. A piece of cloth is 5 m long. A tailor cuts pieces of 50 cm each from it. How many such pieces can he cut?

- (A) 5 (B) 10
(C) 25 (D) 50



Q28. The bar graph shows toffees sold by a shop in four days. On which day were the MOST toffees sold, and how many?



(A) Tuesday, 45

(C) Monday, 30

(B) Thursday, 50

(D) Wednesday, 25

Q29. Rohan weighs 28 kg and his younger brother weighs 19 kg 500 g. What is their total weight?

(A) 46 kg 500 g

(C) 47 kg

(B) 47 kg 500 g

(D) 48 kg 500 g

Q30. A movie show runs 5 times a day. Each show has 120 seats and they are always full. How many people watch the movie in one day?

(A) 500

(C) 625

(B) 600

(D) 720

SECTION 4 · ACHIEVERS SECTION · Q31-Q35 · (2 marks each)

These are the highest-scoring questions. Read each one carefully — they need more than one step.

Q31. A number is such that when you add 1,250 to it and then subtract 680, you get 4,000. What is the number?

(A) 3,430

(C) 2,750

(B) 3,320

(D) 5,430

Q32. Find the value: **(half of 48) + (one-third of 27) – (one-quarter of 20).**

(A) 24

(C) 30

(B) 28

(D) 33

Q33. A rectangular floor is 6 m long and 4 m wide. Square tiles of side 1 m are used to cover it fully. If each tile costs ₹35, what is the total cost of the tiles?

(A) ₹700

(C) ₹350

(B) ₹840

(D) ₹240

Q34. Today is Wednesday. What day of the week will it be after **100 days**?

(A) Thursday

(C) Saturday

(B) Friday

(D) Sunday



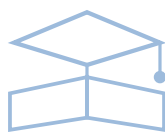
Q35. In a box there are some red and blue balls. The number of red balls is 3 times the number of blue balls. If there are 48 balls in all, how many are RED?

(A) 12

(B) 16

(C) 24

(D) 36





ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	A	A	B	D	D	C	B	A	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	D	A	A	B	C	B	A	C	C	C
Q	21	22	23	24	25	26	27	28	29	30
Ans	C	B	B	A	B	A	B	B	B	B
Q	31	32	33	34	35					
Ans	A	B	B	B	D					

Step-by-Step Solutions

Q1. Correct answer: (B)

The differences are 4, 6, 8, 10, ... so the next difference is 12. $30 + 12 = 42$. (Also, each term = $n \times (n+1)$: 1×2 , 2×3 , 3×4 , ... so the 6th term is $6 \times 7 = 42$.)

Q2. Correct answer: (A)

Move each letter one step forward: $F \rightarrow G$, $R \rightarrow S$, $I \rightarrow J$, $D \rightarrow E$, $A \rightarrow B$, $Y \rightarrow Z$. So FRIDAY $\rightarrow$ **GSJEBZ**.

Q3. Correct answer: (A)

'My mother's only son' is Aarti's brother — but if Aarti has no brother, then 'my mother's only son' is Aarti herself. The son of Aarti is her **son**. (Trick: read 'only son' carefully — it points back to Aarti.)

Q4. Correct answer: (B)

The dots increase by 2 each time: 1, 3, 5, 7, so the next box has **9 dots**.

Q5. Correct answer: (D)

Order from shortest: Sita < Reena < Meena < Gita. So the tallest is **Gita**. Write the names in a line from short to tall to compare easily.

Q6. Correct answer: (D)

A square, rectangle and triangle are all made of straight lines (they have corners and sides). A **circle** has no straight sides or corners, so it is the odd one out.

Q7. Correct answer: (C)

At 3 o'clock the minute hand points to 12 and the hour hand to 3. There are 3 hour-gaps between them, and each hour-gap is 30° , so $3 \times 30^\circ = 90^\circ$ (a right angle).

Q8. Correct answer: (B)

$2 \times 4 = 8$, so the rule is 'multiply by 4'. Then $3 \times 4 = 12$.

Q9. Correct answer: (A)

Starting North and turning right means he now faces **East**. He makes a right angle: 4 km North and 3 km East form a 3-4-5 right triangle, so the straight-line distance is **5 km**.

Q10. Correct answer: (C)

Position from right = (total — position from left) + 1 = $(20 - 7) + 1 = 14\text{th}$. Remember to add 1 because Anu herself is counted.

**Q11. Correct answer: (D)**

In 47,365 the digit 7 is in the thousands place, so its place value is **7,000**.

Q12. Correct answer: (A)

$6,408 = 6 \text{ thousands} + 4 \text{ hundreds} + 0 \text{ tens} + 8 \text{ ones} = \mathbf{6,000 + 400 + 8}$. There are no tens, so we skip them.

Q13. Correct answer: (A)

Do $\div$ and $\times$ before $+$. $72 \div 8 = 9$ and $5 \times 3 = 15$. Then $9 + 15 = \mathbf{24}$. Always do divide and multiply first.

Q14. Correct answer: (B)

$3/6 = 3 \div 3 \text{ over } 6 \div 3 = 1/2$. So **$3/6$** is equal to one half. ($4/6$ simplifies to $2/3$, not $1/2$.)

Q15. Correct answer: (C)

A square can be folded so the two halves match exactly in **4** ways: two through the middle of opposite sides, and two along the diagonals.

**Q16. Correct answer: (B)**

We need a number that gives 56 when multiplied by 8. Since $8 \times 7 = 56$, the missing number is **7**.

Q17. Correct answer: (A)

There are 10 equal squares in all and 4 are shaded, so the shaded fraction is **$4/10$** (which can also be written as $2/5$).

Q18. Correct answer: (C)

A square has 4 equal sides, so perimeter $= 4 \times \text{side} = 4 \times 9 = \mathbf{36 \text{ cm}}$. (81 cm^2 would be its area.)

Q19. Correct answer: (C)

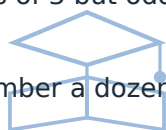
To round to the nearest thousand, look at the hundreds digit. Here it is 5, which means round up, so 3,567 becomes **4,000**.

Q20. Correct answer: (C)

A multiple of 3 that is also even must be in the table of 3 and end in an even digit. $18 = 3 \times 6$ and is even, so **18** fits both. (9 and 15 are multiples of 3 but odd; 25 is neither.)

Q21. Correct answer: (C)

1 dozen = 12. Cost $= 12 \times ₹6 = \mathbf{₹72}$. Remember a dozen always means 12.

**Q22. Correct answer: (B)**

$4/5 \text{ of } 250 = (250 \div 5) \times 4 = 50 \times 4 = \mathbf{200 \text{ L}}$.

Q23. Correct answer: (B)

From 9:45 a.m. to 12:45 p.m. is 3 hours. From 12:45 to 1:15 p.m. is 30 minutes more. Total = **3 hours 30 minutes**.

Q24. Correct answer: (A)

Total spent $= ₹185 + ₹45 = ₹230$. Money left $= ₹500 - ₹230 = \mathbf{₹270}$.

Q25. Correct answer: (B)

Fencing goes around the boundary, so we need the perimeter $= 2 \times (12 + 8) = 2 \times 20 = \mathbf{40 \text{ m}}$.

Q26. Correct answer: (A)

Total chairs $= 6 \times 15 = 90$. Occupied $= 90 - 18 = \mathbf{72}$. Multiply for the total, then subtract the empty ones.

Q27. Correct answer: (B)

$5 \text{ m} = 500 \text{ cm}$. Number of pieces $= 500 \div 50 = \mathbf{10}$. Always convert to the same unit first.

**Q28. Correct answer: (B)**

The tallest bar is Thursday at 50 toffees, so the most were sold on **Thursday (50)**.

Q29. Correct answer: (B)

28 kg = 28,000 g and 19 kg 500 g = 19,500 g. Total = 47,500 g = **47 kg 500 g**.

Q30. Correct answer: (B)

People per day = 5 shows $\times$ 120 seats = **600**.

Q31. Correct answer: (A)

Work backwards from 4,000. Before subtracting 680, the value was $4,000 + 680 = 4,680$. Before adding 1,250, the number was $4,680 - 1,250 = \mathbf{3,430}$.

Q32. Correct answer: (B)

Half of 48 = 24. One-third of 27 = 9. One-quarter of 20 = 5. So $24 + 9 - 5 = \mathbf{28}$.

Q33. Correct answer: (B)

Number of tiles = area of floor = $6 \times 4 = 24$ square metres = 24 tiles. Cost = $24 \times ₹35 = \mathbf{₹840}$. (Area gives the number of 1 m $\times$ 1 m tiles needed.)

Q34. Correct answer: (B)

Every 7 days the day repeats. $100 \div 7 = 14$ weeks and remainder 2. So count 2 days after Wednesday: Thursday (1), **Friday** (2). The answer is Friday.

Q35. Correct answer: (D)

Think in equal parts: blue = 1 part, red = 3 parts, total = 4 parts = 48 balls. So 1 part = $48 \div 4 = 12$ (blue). Red = 3 parts = $3 \times 12 = \mathbf{36}$.

